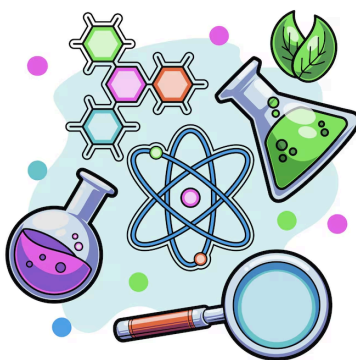




UNIVERSITY INTERSCHOLASTIC LEAGUE

Science

Invitational A • 2026



GENERAL DIRECTIONS:

- **DO NOT OPEN EXAM UNTIL TOLD TO DO SO.**
- Contestants may take up to two hours to complete the contest. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet. Write clearly and legibly!
- You may place as many notations as you desire anywhere on the test paper but not on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- The back two pages of this test include a copy of the periodic table of the elements, as well as listings of other scientific relationships. You may use this information during the contest and may detach the back page from the test if you wish.
- A simple scientific calculator is sufficient for the high school Science contest. **The UIL provides a list of approved calculators that meet the criteria for use in the Science contest. No other calculators are permitted during the contest.** The Science Contest Approved Calculator List is available in the current Science Contest Handbook and on the UIL website. Contest directors will perform a brief visual inspection to confirm that all contestants are using only approved calculators. Each contestant may use up to two approved calculators during the contest.

- B01. Which of the following occurs during anaphase of mitosis?
- A) Chromosomes decondense and nuclear envelopes reform.
 - B) DNA is replicated and the cell prepares for division.
 - C) Spindle fibers attach to centromeres and chromosomes align.
 - D) Chromosomes condense and spindle fibers begin to form.
 - E) Sister chromatids are pulled to opposite poles.
- B02. Phenotype refers to
- A) The number of gametes in an individual.
 - B) An individual's genetic composition.
 - C) The combination of alleles in an individual.
 - D) The observable expression of the genes in an individual.
 - E) The number of chromosomes in an individual.
- B03. _____ reactions release energy by breaking down complex molecules.
- A) Endergonic
 - B) Anabolic
 - C) Catabolic
 - D) Photosynthetic
 - E) Reductive
- B04. Which organ system is primarily responsible for transporting oxygen, nutrients, and waste throughout the body?
- A) Respiratory
 - B) Circulatory
 - C) Digestive
 - D) Nervous
 - E) Integumentary
- B05. Where does transcription of most prokaryotic genes primarily occur?
- A) Inside the nucleus.
 - B) On the rough endoplasmic reticulum.
 - C) Within the mitochondria.
 - D) In the cytoplasm.
 - E) On free ribosomes.
- B06. A population of beetles has two color morphs: green and brown. Birds preferentially prey on green beetles. Over several generations, the frequency of brown beetles increases. Which evolutionary mechanism best explains this change?
- A) Genetic drift
 - B) Gene flow
 - C) Natural selection
 - D) Mutation
 - E) Sexual selection
- B07. If helicase function is blocked during DNA replication, which process would be directly affected?
- A) Separation of parental DNA strands.
 - B) Rate of DNA synthesis.
 - C) Segregation of sister chromatids.
 - D) Transcription initiation at promoters.
 - E) Frequency of spontaneous mutations.
- B08. Which process in the nitrogen cycle converts atmospheric nitrogen (N_2) into ammonia (NH_3), making nitrogen available to living organisms?
- A) Nitrogen fixation
 - B) Denitrification
 - C) Ammonification
 - D) Nitrification
 - E) Assimilation

- B09. In November 2025, the Centers for Disease Control and Prevention announced a multistate outbreak of *Listeria* linked to
- A) peaches.
 - B) cookie dough.
 - C) pistachio ice cream.
 - D) Prepared pasta meals.
 - E) leafy greens.
- B10. During dehydration (condensation) synthesis, which of the following occurs?
- A) Monomers are broken apart by adding a molecule of water.
 - B) Water is used to cleave covalent bonds.
 - C) Covalent bonds are broken by hydrolytic enzymes.
 - D) Polymers are joined together by removing a molecule of water.
 - E) Monomers are joined together by removing a molecule of water.
- B11. In photosynthesis, if the chloroplasts are damaged, which immediate effect will occur?
- A) The plant will increase glucose production.
 - B) Cellular respiration rates will decrease.
 - C) The plant will be unable to convert light energy into chemical energy.
 - D) The plant will increase oxygen production.
 - E) The plant's water loss through transpiration will not change significantly.
- B12. A scientist inserts a human insulin gene into bacteria so the bacteria will produce insulin that can be harvested for medical use. This is an example of:
- A) Genetic engineering
 - B) Selective breeding
 - C) Mutation
 - D) Natural selection
 - E) Gene therapy
- B13. Two organisms belonging to the same Class would also belong to the same ____.
- A) Family
 - B) Order
 - C) Genus
 - D) Phylum
 - E) Order
- B14. In a study of human anatomy, scientists discovered a new, discrete structure that carries out specific functions (like filtering waste). To properly classify this structure, which term should they use, and why?
- A) Organ system — because it consists of multiple organs coordinating functions
 - B) Organ — because it is made of various tissues working as a functional unit
 - C) Gene — because it controls the development of tissues
 - D) Tissue — because it is a group of similar cells performing the same function
 - E) Molecule — because it is the smallest unit of biological function
- B15. Which of the following best describes commensalism in an ecological community?
- A) Both species benefit from the interaction.
 - B) One species benefits while the other is harmed.
 - C) Both species are harmed.
 - D) One species uses a resource without affecting its availability to others.
 - E) One species benefits while the other is unaffected.
- B16. Which of the following is the causative agent of whooping cough?
- A) *Mycoplasma pneumoniae*
 - B) *Bordetella pertussis*
 - C) *Haemophilus influenzae*
 - D) *Streptococcus pyogenes*
 - E) *Legionella pneumophila*

- B17. A patient suffers an injury that damages the tissue responsible for voluntary movement. Which tissue type was most likely injured?
- A) Cardiac tissue
 - B) Skeletal muscle tissue
 - C) Smooth muscle tissue
 - D) Nervous tissue
 - E) Connective tissue
- B18. Which of the following processes requires energy input from the cell to move substances across the plasma membrane?
- A) Osmosis of water.
 - B) Diffusion of oxygen.
 - C) Facilitated diffusion of glucose.
 - D) Active transport of sodium ions.
 - E) Movement of lipids across the bilayer.
- B19. A biologist finds two populations of frogs that live in the same area but breed at different times of year. Which form of reproductive isolation is likely preventing gene flow between them?
- A) Geographic isolation
 - B) Behavioral isolation
 - C) Temporal isolation
 - D) Mechanical isolation
 - E) Hybrid inviability
- B20. Which of the following best describes a silent mutation?
- A) A mutation that causes a stop codon to appear early, ending translation prematurely.
 - B) A mutation that changes an amino acid in the protein sequence.
 - C) A mutation that does not change the amino acid specified by the codon.
 - D) A mutation that deletes an entire gene.
 - E) A mutation that shifts the reading frame, altering every codon after it.

C01. How many atoms are there in a nitrate ion?

- A) 1 B) 2 C) 3 D) 4 E) 5

C02. What are the products of the reaction when solid aluminum hydroxide, $\text{Al}(\text{OH})_3$, is reacted with aqueous hydrochloric acid, HCl ?

- A) H_2O , Al , and Cl_2
 B) H_2O and AlCl_3
 C) H^+ , OH^- , and AlCl
 D) H_2 , O_2 , and AlCl_3
 E) H_2O , H_2 , and AlCl_3

C03. The guideline that says atoms and ions tend to be stable when their valence shell contains exactly eight electrons is known as

- A) Kinderhook's Law
 B) Henry the Eighth's Law
 C) Ocho Osos
 D) The Octet Rule
 E) The Rule of Eight

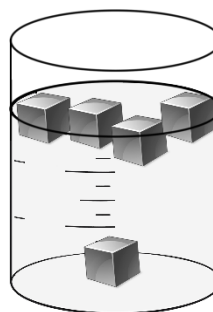
C04. A covalent double bond such as the one found between the carbon atom and each oxygen atom in a carbon dioxide molecule contains

- A) 2 σ bonds
 B) 2 π bonds
 C) 1 σ bond and 1 π bond
 D) 2 σ bonds and 1 π anti-bond
 E) 1 σ anti-bond and 2 π bonds

C05. What is the mass of helium gas in a fully inflated standard-sized 11" party balloon (call it 14.0 L) at 25°C and 1 atm?

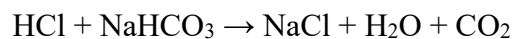
- A) 2.29 g
 B) 2.50 g
 C) 4.00 g
 D) 0.625 g
 E) 0.573 g

C06. The five cubes shown in this diagram are made of copper, lead, tungsten, aluminum, and steel, and they have been dropped into a container of liquid mercury. Which cube is the one at the bottom?



- A) Copper
 B) Lead
 C) Tungsten
 D) Aluminum
 E) Steel

C07. What is the sum of the coefficients when the reaction between hydrochloric acid and sodium bicarbonate is balanced using the smallest whole numbers?

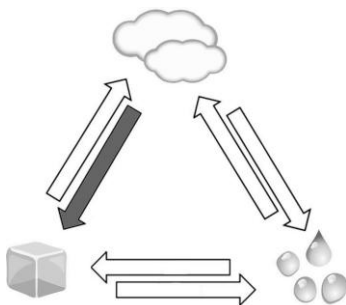


- A) 6 B) 7 C) 8 D) 4 E) 5

C08. If you add 10.0 grams of solid NaCl to a 1 L solution that is saturated in aqueous NaCl , how much solid NaCl will remain undissolved on the bottom of the container? (Ignore any temperature or volume changes due to adding the solid NaCl .)

- A) 10.0 g – none of the added NaCl will dissolve
 B) 0.0 g – the added NaCl will dissolve
 C) less than 10.0 g, but more than 0 g – some of the added NaCl will dissolve, but because the solution is already saturated not all of it will dissolve
 D) more than 10.0 g – the added NaCl will cause some of the dissolved NaCl to precipitate out of the solution.
 E) none of the above answers are correct

C09. What is the name of the phase change indicated by the solid arrow?



- A) Sublimation
- B) Deposition
- C) Solidification
- D) Fusion
- E) Condensation

C10. In a chemical reaction that has reached equilibrium

- A) the reaction has stopped
- B) the reaction continues in a net forward direction, but slowly
- C) the moles of products are equal to the amount of reactants
- D) the concentration of the products is equal to the concentration of the reactants
- E) none of the above answers are correct

C11. For the dissociation of a generic weak acid HA, the ionization constant K_a is equal to

- A) $[HA]$
- B) $[H^+][A^-]$
- C) $\frac{[H^+][OH^-]}{[HA]}$
- D) $\frac{[H^+][A^-]}{[HA]}$
- E) $[HA][H^+][OH^-]$

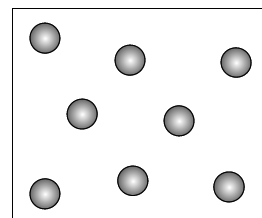
C12. When an atom undergoes oxidation, the atom

- A) gains electrons, leading to a higher oxidation number
- B) loses electrons, leading to a lower oxidation number
- C) gains electrons, leading to a lower oxidation number
- D) loses electrons, leading to a higher oxidation number
- E) becomes an oxygen atom

C13. If you want to determine the instantaneous rate of a chemical reaction at any given point in time, which of these would you use to calculate it?

- A) the equilibrium expression for the reaction
- B) the law of mass action for the equation
- C) the rate law for the reaction
- D) the integrated rate law for the reaction
- E) the Arrhenius equation

C14. In this diagram of a gas sample, each sphere represents one atom. Which of these gases could it be?

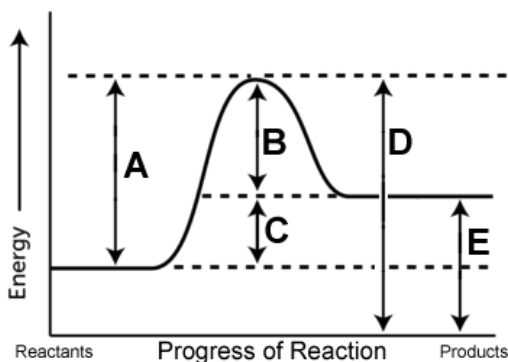


- A) Nitrogen
- B) Argon
- C) Oxygen
- D) Carbon Dioxide
- E) Fluorine

C15. A partial neutralization reaction is carried out by adding 230 mL of 0.025 M NaOH to 120 mL of 0.050 M HCl. What is the pH of the resulting solution?

- A) 1.30
- B) 3.15
- C) 5.65
- D) 7.00
- E) 9.65

- C16. In the potential energy diagram shown below, which arrow corresponds to the activation energy for the forward reaction?



- A) A
B) B
C) C
D) D
E) E
- C17. Given this word equation:
“Ethylene glycol reacts with oxygen to produce carbon dioxide and water.”
What type of reaction is this?
A) Acid-base
B) Synthesis
C) Combustion
D) Double Replacement
E) Single Replacement
- C18. Which of these compounds has the highest percent copper by mass?
A) CuO
B) CuCl₂
C) CuSO₄
D) Cu(NO₃)₂
E) CuS
- C19. What is the solid product formed when aqueous Ba(NO₃)₂ is mixed with aqueous NaBrO₃?
A) barium nitrate
B) barium oxide
C) barium bromide
D) barium bromite
E) barium bromate
- C20. How many coulombs of charge are necessary to reduce 0.200 mol of Ag⁺ ions to solid Ag?
A) 19,297 C
B) 96,485 C
C) 107.87 C
D) 539.35 C
E) 1.20×10^{23} C

- P01. According to Malley, in 1895, Wilhelm Röntgen found a new type of invisible radiation which could penetrate solid, opaque objects. What did Röntgen discover?
- Infrared radiation
 - Gamma rays
 - Beta particles
 - X-rays
 - Neutrinos
- P02. According to Malley, early in her career, Marie Sklodowska (Curie) studied _____ for the Society for the Encouragement of National Industry.
- the magnetic properties of steel
 - the alloying properties of aluminum
 - the electrical properties of iridium
 - the physical properties of tungsten
 - the radioactive properties of radium
- P03. According to Malley, Pierre Curie and his brother Jacques discovered a material that gave off electrical signals when compressed. This is known as the piezo-electric effect. With what material did the Curies discover this effect?
- diamonds
 - garnets
 - thorium crystals
 - uranium crystals
 - quartz crystals
- P04. A star that is powered by hydrogen fusion in its core and is about midway through its lifetime is classified as a
- regular type star
 - main sequence star
 - common sequence star
 - ordinary star
 - proto star
- P05. For the calculation shown, which answer choice has the correct number of significant figures?
- $$X = \frac{11.31 - 6.403}{1.2188}$$
- $X = 4$
 - $X = 4.0$
 - $X = 4.03$
 - $X = 4.026$
 - $X = 4.0261$
- P06. You startle a squirrel, which accelerates horizontally from rest at a constant rate of 6.65m/s^2 . How long does it take for the squirrel to reach its maximum speed of 11.4m/s ?
- 0.857 sec.
 - 1.31 sec.
 - 1.71 sec.
 - 2.62 sec.
 - 3.43 sec.
- P07. Two horizontal forces act on a box of dog food, as shown. The mass of the box of food is 5.75kg . What is the magnitude of the acceleration of the box of food?
- 0.870 m/s^2
 - 2.26 m/s^2
 - 3.13 m/s^2
 - 3.86 m/s^2
 - 5.39 m/s^2
- 
- P08. You use a fixed vertical spring to launch a rock straight up into the air. The spring constant of the spring is 240N/m , and the rock's mass is 42.0g . You compress the spring by 5.50cm , place the rock on it, and release the spring. What is the velocity of the rock at the instant that it launches from the spring?
- 2.94 m/s
 - 4.03 m/s
 - 5.88 m/s
 - 16.2 m/s
 - 31.4 m/s
- P09. A meter stick is placed horizontally with a fulcrum set at the 50.0cm mark. A coin with an unknown mass is attached to the meter stick at the 80.0cm mark. A known mass of 130.0g is attached to the meter stick and its position is adjusted to balance the unknown mass. The system balances, and is in perfect equilibrium, when the known mass is located at the 10.7cm mark on the meter stick. What is the mass of the coin?
- 17.4 g
 - 46.4 g
 - 62.9 g
 - 170 g
 - 364 g

P10. You own a pendulum that has a period of oscillation of 1.50sec on Earth. What would be the period of oscillation of this same pendulum if you were to swing it on the Moon? [Note: the acceleration due to gravity on the Moon is 1.62m/s^2]

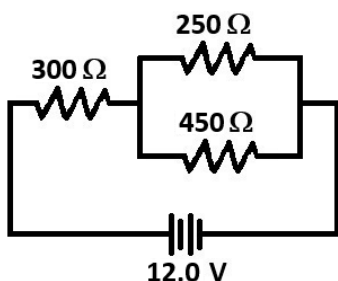
- A) 0.248 sec.
- B) 0.610 sec
- C) 1.50 sec.
- D) 3.70 sec.
- E) 9.07 sec.

P11. A small stone has a mass of 117.0g. When placed in a graduated cylinder containing 60.0ml of water, the stone causes the water level to rise to 96.0ml. What is the density of the stone?

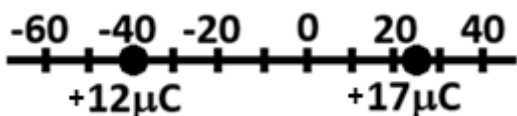
- A) 750 kg/m^3
- B) 1220 kg/m^3
- C) 1950 kg/m^3
- D) 2600 kg/m^3
- E) 3250 kg/m^3

P12. Determine the current flowing in the 450Ω resistor in the circuit shown.

- A) 9.3 mA
- B) 14 mA
- C) 17 mA
- D) 26 mA
- E) 40 mA



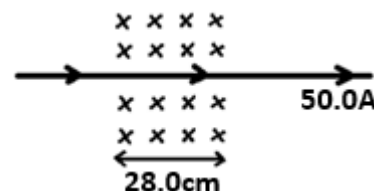
P13. Two charges are placed as shown. A $+12.0\mu\text{C}$ charge is located at $x = -40.0\text{cm}$, and a $+17.0\mu\text{C}$ charge is located at $x = +25.0\text{cm}$. What is the magnitude of the force on the $+17.0\mu\text{C}$ charge due to the $+12.0\mu\text{C}$ charge?



- A) 1.83 N
- B) 2.82 N
- C) 4.34 N
- D) 7.34 N
- E) 11.5 N

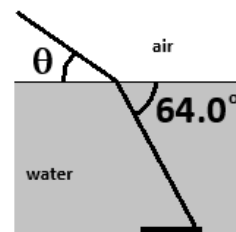
P14. A long straight wire carries a current of 50.0A, as shown. The wire passes through a region with a 400.0 Gauss magnetic field, oriented perpendicular to the wire. The region of magnetic field is 28.0cm wide. What is the net magnetic force acting on the current carrying wire?

- A) 0.56 N
- B) 2.0 N
- C) 7.1 N
- D) 14 N
- E) 35 N



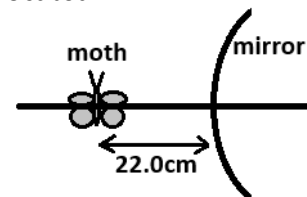
P15. Sadly, your cell phone has been dropped into a lake. A beam of light from your phone shines as illustrated. At what angle, θ , relative to the surface of the water, does the beam of light travel? The index of refraction of the lake water is 1.33.

- A) 35.7°
- B) 42.5°
- C) 47.5°
- D) 54.3°
- E) 64.0°



P16. A convex mirror is placed in a convenience store to give the clerk a good view of the entire store. The magnitude of the radius of curvature of the mirror is 72.0cm. A moth flutters in front of the mirror, at a distance of 22.0cm from the mirror's surface, as shown. How far from the mirror's surface is the virtual image of the moth located?

- A) 9.90 cm
- B) 13.7 cm
- C) 16.9 cm
- D) 31.7 cm
- E) 56.6 cm



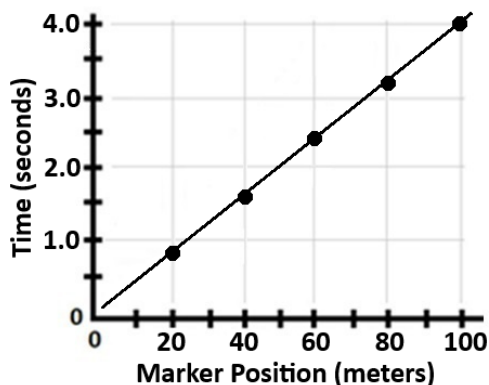
P17. An electron is confined in a quantum dot that has a width of 25.0nm. If we use the size of the quantum dot as the uncertainty in position of the electron, then what is the approximate uncertainty in the velocity of the electron?

- A) $2 \times 10^2\text{ m/s}$
- B) $2 \times 10^3\text{ m/s}$
- C) $2 \times 10^4\text{ m/s}$
- D) $2 \times 10^5\text{ m/s}$
- E) $2 \times 10^6\text{ m/s}$

P18. A meson is discovered that has a mass of about $2000 \text{ MeV}/c^2$, an average lifetime of 5.0×10^{-13} seconds, a charge number of -1 , strangeness of -1 , and charm of -1 . What is the quark structure of this meson?

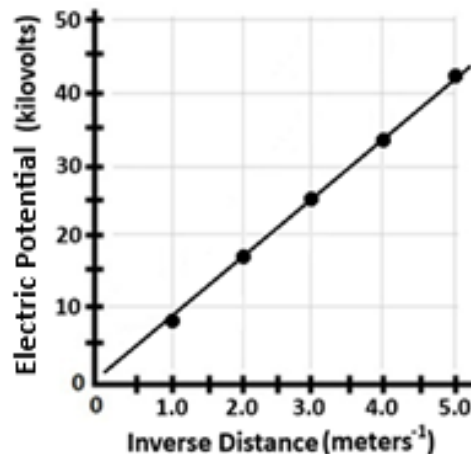
- A) usc
- B) dsc
- C) $\bar{u}s$
- D) $\bar{c}d$
- E) $\bar{c}s$
- F) $c\bar{s}$

P19. The following plot shows the times at which a galloping horse passes fixed position markers along a race track. What is the speed at which the horse is galloping?



- A) 85 m/s
- B) 70 m/s
- C) 55 m/s
- D) 40 m/s
- E) 25 m/s

P20. The plot shown below gives the electric potential (V) measured near a point charge, Q . The potential is plotted against the inverse of the distance from the charge ($1/r$). From these data, determine the approximate magnitude of the charge Q .



- A) 600 nC
- B) 800 nC
- C) 950 nC
- D) 1100 nC
- E) 1300 nC

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

T_{mp}	$= 0^{\circ}\text{C}$
T_{bp}	$= 100^{\circ}\text{C}$
c_{ice}	$= 2.09 \text{ J/g}\cdot\text{K}$
c_{water}	$= 4.184 \text{ J/g}\cdot\text{K}$
c_{steam}	$= 2.03 \text{ J/g}\cdot\text{K}$
ΔH_{fus}	$= 334 \text{ J/g}$
ΔH_{vap}	$= 2260 \text{ J/g}$
K_{f}	$= 1.86^{\circ}\text{C}/m$
K_{b}	$= 0.512^{\circ}\text{C}/m$

$$\begin{aligned} R &= 0.08206 \text{ L}\cdot\text{atm/mol}\cdot\text{K} \\ R &= 8.314 \text{ J/mol}\cdot\text{K} \\ R &= 62.36 \text{ L}\cdot\text{torr/mol}\cdot\text{K} \\ e &= 1.602 \times 10^{-19} \text{ C} \\ N_{\text{A}} &= 6.022 \times 10^{23} \text{ mol}^{-1} \\ k &= 1.38 \times 10^{-23} \text{ J/K} \\ h &= 6.626 \times 10^{-34} \text{ J}\cdot\text{s} \\ c &= 3.00 \times 10^8 \text{ m/s} \\ R_{\text{H}} &= 2.178 \times 10^{-18} \text{ J} \\ m_{\text{e}} &= 9.11 \times 10^{-31} \text{ kg} \\ \mathcal{F} &= 96,485 \text{ C/mol e}^{-} \\ 1 \text{ amp} &= 1 \text{ C/sec} \\ 1 \text{ mol e}^{-} &= 96,485 \text{ C} \end{aligned}$$

Copper	8.96
Aluminum	2.71
Mercury	13.6
Tungsten	19.3
Osmium	22.6
Steel	7.85
Lead	11.3
Platinum	21.5

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Physics Useful Constants

quantity	symbol	value
Free-fall acceleration	g	9.80 m/s^2
Permittivity of Free Space	ϵ_0	$8.854 \times 10^{-12} \text{ C}^2/\text{Nm}^2$
Permeability of Free Space	μ_0	$4\pi \times 10^{-7} \text{ Tm/A}$
Coulomb constant	k	$8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$
Speed of light in a vacuum	c	$3.00 \times 10^8 \text{ m/s}$
Fundamental charge	e	$1.602 \times 10^{-19} \text{ C}$
Planck's constant	h	$6.626 \times 10^{-34} \text{ Js}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67265 \times 10^{-27} \text{ kg}$ 1.007276 amu
Neutron mass	m_n	$1.67495 \times 10^{-27} \text{ kg}$ 1.008665 amu
Atomic Mass Unit	amu	$1.66 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
Gravitational constant	G	$6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
Stefan-Boltzmann constant	σ	$5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$
Universal gas constant	R	$8.314 \text{ J/mol} \cdot \text{K}$ $0.082057 \text{ L} \cdot \text{atm/mol} \cdot \text{K}$
Boltzmann's constant	k_B	$1.38 \times 10^{-23} \text{ J/K}$
Speed of Sound (at 20°C)	v	343 m/s
Avogadro's number	N_A	$6.022 \times 10^{23} \text{ atoms/mol}$
Electron Volts	eV	$1.602 \times 10^{-19} \text{ J/eV}$
Distance Conversion	miles $\rightarrow$ meters inches $\rightarrow$ centimeters	1.0 mile = 1609 meters 1.00 inch = 2.54 centimeters
Rydberg Constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
Standard Atmospheric Pressure	1 atm	$1.013 \times 10^5 \text{ Pa}$
Density of Pure Water	ρ_{water}	1000.0 kg/m^3
Magnetic Field Conversion	Gauss $\rightarrow$ Tesla	$10^4 \text{ Gauss} = 1.00 \text{ Tesla}$